UrbanLens — Milestone 2 Report

Real Data Ingestion + Python Pipeline + Express APIs

> IMPORTANT
>
> All suites passing on actual hardware:
>
> - **Python tests:** 25 / 25 ✅
> - **Pipeline smoke test:** 2 / 2 datasets ✅
> - **Server API tests:** 10 / 10 ✅
> - **Client build:** `tsc -b && vite build` → exit 0 ✅

Git Checkpoints

Commit	Message
`c0e4d61`	`feat: milestone 2 — data pipeline, Express APIs, frontend integration`
`7183953`	`fix: resolve pandas Series.copy dtype coercion in validators (test_04)`

Commands to Run the Application

```
bash
cd "/Users/vanshbarnes/Documents/UrbanLens Project"

# Frontend (http://localhost:5173)
npm --prefix client run dev

# Backend API (http://localhost:3001)
npm --prefix server run dev

# Python pipeline — synthetic fixtures (no real data needed)
cd server/python && python3 run_pipeline.py --test

# Ingest a real CSV (after manual download)
cd server/python
python3 run_pipeline.py \
  --dataset hpi \
  --file data/raw/rbi_hpi_raw.csv \
  --source-name RBI-HPI \
  --source-url https://data.rbi.org.in \
  --status real

# Fetch OSM amenity data for Mumbai (requires internet)
python3 osm_fetch.py \
  --city Mumbai \
  --bbox "18.89,72.77,19.27,72.99" \
  --output data/raw/osm_mumbai_amenities.csv

# Run all tests
cd server/python && python3 -m unittest discover tests -v
cd server && npm test
cd client && npm run build
```

Test Results (Independently Verified)

Python Pipeline — 25/25 ✅

```
test_01_valid_hpi_fixture_ingestion           ok
test_02_invalid_csv_missing_columns           ok
test_03_invalid_date_format_warning_rejection ok
test_04_invalid_metric_value_rejection        ok  ← fixed here
test_05_duplicate_detection_and_removal       ok
test_06_metric_type_consistency_warning       ok
test_07_quality_report_structure              ok
test_08_normalizer_indian_fy_quarter_to_iso   ok
test_09_geography_name_normalization          ok
test_10_amenities_csv_validation              ok
test_11_empty_csv_handling                    ok
+ 14 unit tests for validators/normalizers/quality
Ran 25 tests in 0.066s — OK
```

Pipeline Smoke Test ✅

```
[1/2] HPI fixture  → 13 valid records, 1 duplicate removed, cities: Delhi, Mumbai
[2/2] Amenities    → 4 valid records, 1 duplicate removed, types: hospital, park,
school
Output: server/python/data/processed/
```

Express API Tests — 10/10 ✅

```
✔ GET /api/health → 200 ok
✔ GET /api/data/sources → 200 DataSource[]
✔ GET /api/data/series → 200 observations[]
✔ GET /api/data/series?start_date=bad → 400
✔ GET /api/data/series?geography=../../etc/passwd → 400 (path traversal blocked)
✔ GET /api/data/quality → 200
✔ GET /api/data/summary → 200
✔ GET /api/neighborhoods/amenities → 200
✔ GET /api/neighborhoods/amenities?amenity_type=school → 200
✔ start_date > end_date → 400
```

Client Build ✅

```
✓ 2267 modules transformed · built in 1.94s · exit 0
```

Files Created / Modified

Python Pipeline (`server/python/`)

File	Role
`requirements.txt`	`pandas>=2.0` , `numpy>=1.24`
`pipeline/schema.py`	Schema constants + HPI-vs-price separation rules
`pipeline/validators.py`	Column/date/type/dedup validation
`pipeline/normalizers.py`	Indian FY quarter → ISO date, geography aliases
`pipeline/quality.py`	Machine-readable quality report
`pipeline/ingest.py`	Orchestrator: read → validate → normalize → dedup → save
`run_pipeline.py`	CLI entry (`--test` or `--dataset/--file`)
`osm_fetch.py`	Standalone OSM Overpass fetcher (stdlib only)
`data/raw/README.md`	Manual browser-export workflow
`data/fixtures/test_hpi_fixture.csv`	SYNTHETIC_TEST_DATA, 14 rows, 1 duplicate
`data/fixtures/test_amenities_fixture.csv`	SYNTHETIC_TEST_DATA, 5 rows, 1 duplicate
`tests/test_ingestion.py`	11 end-to-end tests
`tests/test_validators.py`	9 unit tests
`tests/test_quality.py`	Quality report structure test

Express Server (`server/src/`)

File	Role
`types/api.ts`	TypeScript interfaces for all API shapes
`middleware/validateQuery.ts`	Date regex, path-traversal sanitization, range ordering
`services/dataService.ts`	Controlled read-only access over `python/data/processed/`
`routes/health.ts`	`/api/health` v0.2.0
`routes/data.ts`	`/api/data/sources` , `series` , `quality` , `summary`
`routes/neighborhoods.ts`	`/api/neighborhoods/amenities`
`src/index.ts`	Updated — routes + global error handler
`src/__tests__/api.test.ts`	10 API tests (no port binding)
`jest.config.ts`	Jest + ts-jest

Client (`client/src/`)

File	Role
`api/types.ts`	Mirror of server types
`api/client.ts`	Typed fetch wrapper
`api/hooks.ts`	`useDataSources` , `useDataSeries` , `useDataQuality` , `useApiSummary` , `useAmenities`
`components/ui/DataStatusBadge.tsx`	🟢 REAL / 🟡 SYNTHETIC / ⚫ UNAVAILABLE
`components/ui/ApiErrorState.tsx`	Error + retry
`components/ui/DataSourceCard.tsx`	Source metadata card
`pages/DataSources.tsx`	**Updated —** live API, quality reports
`pages/Overview.tsx`	**Updated —** Data Pipeline Status bar

Docs

File	Role
`docs/DATA_SOURCES.md`	Provenance, licensing, coverage, limitations
`README.md`	Milestone 2 section appended

Data Provenance

Source	Status	Unit	License
RBI DBIE HPI	Investigated — manual workflow ready	Index points (base=100), NOT ₹/sqft	GoI Open Data, attribution required
NHB RESIDEX	Investigated — manual workflow ready	Index points	NHB, attribution required
OpenStreetMap	Fetcher script ready	POI counts	© OSM contributors, ODbL
SYNTHETIC_TEST_DATA	**Integrated** — bundled fixtures	N/A — never presented as real	Internal

Bug Fixed in Verification

`test_04_invalid_metric_value_rejection` — ERROR

`pandas.Series.copy()` preserves the source column's `StringDtype`. Writing a `float` back into a string-typed Series raises `TypeError` on pandas ≥ 2.0.

Fix: `row.to_dict()` instead of `row.copy()` — rows accumulated as plain dicts, then assembled into a fresh DataFrame with properly inferred dtypes.

Critical Invariants Enforced

> WARNING

> - HPI index values (dimensionless, base=100) are **never** mixed with ₹/sqft prices
> - City-level HPI does **not** imply neighborhood-level predictions
> - Missing real data → empty array + `data_status: 'unavailable'` — no silent synthetic substitution
> - Path traversal in query params → 400 error (blocked in `validateQuery.ts`)
> - Stack traces never exposed to API clients

Manual Data Import Workflow

1. **Download** from `https://data.rbi.org.in` (Statistics → Real Sector → Prices & Wages → House Price Index → Download CSV)
2. **Normalize columns** to schema: `observation_date, geography_name, geography_level, metric_name, metric_value, unit, source_name, source_url, retrieved_at, data_status`
3. **Place** file in `server/python/data/raw/rbi_hpi_raw.csv`
4. **Run:** `python3 run_pipeline.py --dataset hpi --file data/raw/rbi_hpi_raw.csv --source-name RBI-HPI --source-url https://data.rbi.org.in --status real`
5. **Check** quality report: `cat data/manifests/*.json | python3 -m json.tool`
6. **Reload** — Express `dataService` reads processed files per-request, no restart needed

Known Limitations

Limitation	Notes
No neighborhood-level price data for India	City-level HPI only from official sources
RBI/NHB require manual browser export	No public API or direct CSV URL
OSM fetch requires internet access	Run `osm_fetch.py` manually
JS bundle ~900 kB (no code splitting)	Milestone 3: dynamic imports
`pandas` must be installed separately	`pip3 install -r server/python/requirements.txt`

Recommended Next: Milestone 3 — Forecasting Models

1. Python SARIMA/Prophet forecasting service on ingested HPI data
2. `GET /api/forecasting/predict` endpoint
3. Update Forecasting page — real model confidence intervals
4. Model metadata: training date, MAPE on holdout, prominently labeled as illustrative